Anteater Poker User Manual

Version 1.0 Alpha Draft

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EECS 22L

Note: This manual describes the expected user experience, installation process, game rules, graphical interface, and major functions for the Anteater Poker application. No real money or gambling is involved. The game uses points only and is designed as a friendly online poker game for UCI students.

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Introduction

Anteater Poker is an online multiplayer poker application developed in C. The application is based on the Texas Hold'em style of poker but features unique Anteater-themed mechanics that distinguishes the game from traditional poker.

The purpose of the application is to allow multiple users to connect to a host server and participate in an interactive card game through a graphical user interface (GUI). Players can compete against other human opponents or computer-controlled bots over a network using socket communication.

Anteater Poker consists of a central server application and a graphical user interface. The central server maintains the game state, controls the rounds, shuffles and deals the cards, validates player actions (call, fold, raise), tracks points, and determines the winner of each round. The graphical user interface enables users to connect to the server, interact with the game, and participate in the online gameplay.

The project aims to demonstrate networking, graphical user interface development, modular software design, and software engineering principles through the implementation of an online multiplayer poker experience.

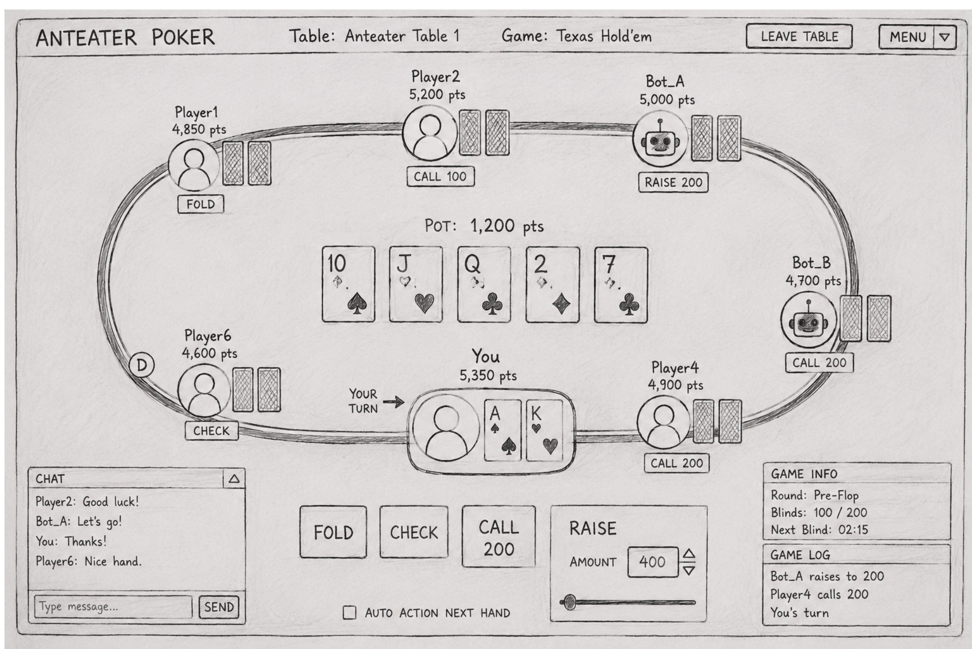
Glossary

- GUI - Graphical User Interface
- TCP/IP - Communication protocol used for networking
- Socket - Communication en dpoint between programs
- Client - Program used by a player to connect to the server
- Server - Central application that hosts the game
- Bot - AI-controlled poker player
- Community Cards - Shared cards used by all players
- Dealer - Player responsible for starting each round
- Anteater Wild Card - A special card that may represent any rank or suit
- Fold - To leave the current betting round
- Raise - To increase the current bet
- Call - To match the current bet
- Check - To pass a turn without betting

1. Online Poker

1.1 Usage Scenario

- Multiple users launch the client application
- Players enter username and select seat
- The server distributes the cards and the players choose to act
- Community cards are revealed, the server determines the winning hand, and then the points are distributed.



1.2 Goals

- To create a multiplayer online poker game
- Provide a clear and active Poker interface
- Ensure everything runs smoothly
- Create a unique Anteater themed variation of poker

1.3 Features

- Multiplayer Support
- GUI Interface - GTK based interface
- Anteater Wild Card - Unique Poker mechanic
- Bot Players - Ai controlled poker players
- Seat Selection
- Dealer Button Rotation
- Server Dashboard

2. Installation

2.1 System requirements

- UCI EECS Linux server or compatible Linux machine.
- C compiler such as gcc.
- Make utility for building the project.
- Git for version control and team collaboration.
- TCP/IP network access between the poker client and poker server.
- Graphical display support for running the client GUI and server GUI.

2.2 Setup and configuration

For a binary release, unpack the submitted archive, open the documentation, start the server, and then run the poker client. The expected workflow is:

$ gtar xvzf Poker_V1.0.tar.gz

$ evince poker/doc/Poker_UserManual.pdf

```
$ poker/bin/server &
$ poker/bin/poker
```

For a source release, unpack the source archive, build the project, run tests, and then start the programs:

```
$ gtar xvzf Poker_V1.0_src.tar.gz
$ cd poker
$ make
$ make test
$ make clean
```

Configuration settings should be stored in a small configuration file or entered on the login screen. Typical settings include server hostname, port number, player display name, and table number.

2.3 Uninstalling

To uninstall the application, close the poker client and server. Then delete the unpacked poker directory from the local machine or team account. If configuration files were saved in the user's home directory, remove those files as well.

```
$ pkill server
```

3. Poker Program Functions and Features

3.1 Client and Server Communication

Anteater Poker uses a client-server system. The server is the main program that hosts the poker table and controls the game. Each player uses a client program to connect to the server. The client sends the player’s actions to the server, such as joining a table, selecting a seat, calling, raising, checking, or folding.

The server receives these actions, checks if they are valid, updates the game state, and sends the new information back to all connected clients. This keeps every player’s screen updated with the same cards, points, turns, and game results. TCP/IP socket communication is used so that the client and server can exchange messages over the network.

The server also handles connection problems. If a player disconnects, the server can notify the other players and allow the disconnected player to reconnect if possible.

3.2 Dealer Choice and Card Distribution

At the beginning of the game, the server assigns a dealer position. The dealer button rotates after each round so that every player gets a fair turn in the dealer position. The server controls this process automatically.

Before each hand, the server shuffles the deck and deals two private cards to each active player. These cards are only visible to the player who receives them. The server then reveals five community cards in stages, following Texas Hold’em rules.

The card distribution happens in this order. First, each player receives two private cards. Next, three community cards are revealed as the flop. Then, one community card is revealed as the turn. After that, one final community card is revealed as the river. Finally, the remaining players enter the showdown.

The server makes sure that cards are not repeated and that all cards are dealt fairly.

3.3 Poker Game Integration

Anteater Poker is based on Texas Hold'em poker. Each player tries to make the best five-card poker hand using their two private cards and the five shared community cards. The game includes common poker actions such as call, raise, check, and fold.

The server controls the flow of the round. It tracks whose turn it is, how many points each player has, the current bet amount, and the total points in the pot. A player may only choose actions that are allowed during their turn.

At the end of the round, the server compares the remaining players' hands and determines the winner. The winning player receives the points from the pot. If there is a tie, the points are split between the tied players.

3.4 Anteater Wild Card Rule

Anteater Poker includes a unique Anteater Wild Card rule. The Anteater Wild Card is a special extra card added to the normal 52-card deck, making the deck a 53-card deck.

The Anteater Wild Card can represent any rank or suit that helps the player create the strongest possible hand. For example, it may act as an ace, king, queen, or any suit needed to complete a flush, straight, full house, or other winning hand.

If a player wins a round using the Anteater Wild Card as part of their final hand, they receive bonus points in addition to the regular points from the pot.

3.5 Bot Player

Anteater Poker includes bot players that can fill empty seats at the table. A bot is a computer-controlled player that can participate in the game when there are not enough human players.

The bot can perform basic poker actions such as checking, calling, raising, or folding. The bot follows simple decision-making rules based on its cards, the current bet, and the game situation.

Bots are managed by the server. The server can add bots to empty seats, remove bots when human players join, and include bots in the normal game.

3.6 Graphical User Interface

The client GUI includes a login screen where players can enter their username, server address, and port number. After connecting, players can choose a seat, view their private cards, see the community cards, check their points, and use buttons for actions such as call, raise, check, and fold.

The client interface may also include a chat area, player list, pot display, and game status messages. These features help players understand what is happening during the round.

4. Error Messages

Anteater Poker may show error messages when something goes wrong, such as a connection problem, invalid input, or an action that is not allowed during the game. Each message explains the problem and helps the user know what to do next. The following messages may appear during the game:

- Connection failed: unable to connect to the server
- Username already taken
- Seat Occupied
- Invalid Bet Amount
- Player disconnected
- Server Timeout
- Server Closed

5. Copyright

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End of User Manual